

95864 AI Model Development (Fall 2026)

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ASSIGNMENT 1#: Datasheets for AI Model Development & Experimenting with Knowledge Representation.

Learning objective:

This assignment is designed to *familiarize you with the concept and practice of* knowledge representation, through data and feature engineering, tokenization and embeddings. For this assignment, you will be asked to explore techniques for:

- Analyzing a training / fine-tuning dataset (its properties, their distribution).
- Creating new features from existing dataset properties.
- Executing prototype experiments to learn how tokenization and embedding strategies affect the semantic meaning of training data (thus, AI behavior).
- Interpreting experiments and using the results to inform the design of LLM development strategies.

The objective of teaching you these techniques is to enable you to:

- Understand how properties of data influence AI model behavior.
- Apply scientific methods to decision-making about AI model development.
- Connect data engineering tasks in AI model development to AI model behavior and broader organization and societal impacts.

Toward this learning objective, in this assignment, you will **analyze and transform a dataset through datasheet development, feature engineering, tokenization, and embedding experiments** to understand how **different knowledge representation strategies and data characteristics** can affect **AI model behavior, model performance, and the semantic interpretation of information by machine learning systems**.

Computer, Software, Data & AI Requirements:

Computing, Software, AI and Data Usage:	
General Software, AI & Computer Use: <i>Please use the software, data, AI or compute resources specified in the class technology FAQ according to the CMU guidelines or rules.</i>	Required steps: [A] Decide what technology resource to use for the assignment. [B] If the assignment requires using technology that has CMU requirements (e.g. HuggingFace), please follow the requirement specified in the Classroom Tech FAQ.
AI Usage for Assignment Report: Prohibited.	Please <u>do not</u> use AI to format, write or edit the assignment report. Students will not be graded on spelling or grammar mistakes. Please practice writing and preparing the report using your own skills.

Due Date & Submission Requirements:

Due Date: Please submit your assignment to Canvas by **Friday, September 4, 2026** at **5:00 PM ET**.

Report Format: For your Assignment Report, please use a **Google Document** to respond to the homework assignment questions.

Canvas submission: Please submit a link/url to your Google document for the homework assignment report.

Grant access to the teaching team: In advance of submitting your assignment report, please provide the teaching team with access to the Google document.

Instructions:

Part A: Organizational Scenario & LLM Training Dataset (7')

[] **Organizational Scenario:** Create an organization scenario for your LLM model training research project (e.g. Assignment 2 and final project). Please refer to the "Organizational

scenario” examples published on our course Canvas. Your scenario does not need to look exactly like these scenarios, they are for reference only.

[] **Organization Scenario Description:** Please describe your organizational scenario in the memo you submit for this assignment.

[] **Project Dataset Selection:** Based on the model scenario, choose a dataset to use for this assignment.

*** **Please note:** this is the dataset your team will use to train an AI model in subsequent assignments and for your final project. For this reason, please meet with your final project team to ensure your choice are aligned.

Dataset selection criteria: You may choose or create a dataset using these resources. Please note the risks of each choice.

Data Source	Description	Restrictions	URL
Class Google Drive	there are datasets in the Google Drive Folder for this class:	N/A	
HuggingFace datasets	you may choose a public dataset published to HuggingFace, but please note, the safety of using these datasets is uncertain and you do so at your own risk.	These datasets <u>cannot</u> be used on CMU computer resources or within CMU environments without prior approval from the university. If you decide to use one of these datasets, you'll need to fulfill the homework requirements by creating a table and providing the requested information about the dataset. Please do not upload a copy of the dataset to Canvas, as I'll not be able to view it. If you will need to use a CMU Virtual Machine or computer to fine-tune a small language model on this dataset, for assignment 2, please consider using a different data source or emailing the teaching team.	
UCI Irvine Machine Learning Repository	you may choose a public dataset published to the UC Irvine Machine Learning Repository, but please note, the safety of using these datasets is uncertain and you do so at your own risk.	These datasets cannot be used on CMU computer resources or within CMU environments without prior approval from the university. These datasets <u>cannot</u> be used on CMU computer resources or within CMU environments without prior approval from the university. If you will need to	

		use a CMU Virtual Machine or computer to fine-tune a small language model on this dataset, for assignment 2, please consider using a different data source or emailing the teaching team.	
AI Generated Synthetic dataset	you may choose to create your own dataset by using an AI model to create synthetic data. However, please note, the safety of using AI models for any task is uncertain and you do so at your own risk.	Unless the AI model is on the approved AI model list for this course, you cannot use the AI model on CMU computer resources or within CMU environments without prior approval from the university. Additionally, you must adhere to CMU's data privacy, security or classification policies when using AI models.	

[] AI Model Training Datasheet: Produce a “datasheet” [2] for the dataset by creating a codebook that describes the features in the dataset and producing these descriptive statistics:

[] Existing Feature Descriptions:

[] Features: What features does the dataset have? List all the named features in the dataset and describe them in the table. Write descriptions for the features using your own words (e.g. please do not use AI to describe the features).

[] Data Types: What are the datatypes of the features (e.g. string/text, numeric)? Please list the datatype of the feature next to its description in the table.

[] Information value/utility for AI Model Dev Scenario (*written analysis*): Using your own words, please write what information these features provide us, and describe what you think the utility or value or relevance of this information could be for the AI Model Development scenario.

[] Data Classification Labels: Using the Traffic Light Protocol [1], create labels for the examples or instances in the dataset. **These labels are now features in your dataset!** Create a table listing and describing each label, and create a third column that provides an example from your dataset.

[] Descriptive Statistics and Data Profiling: Calculate and report descriptive statistics for the features in your dataset. At a minimum, include:

[] Number of observations (instances/examples).

[] Number of features (variables).

- [] Missing values per feature.
- [] Data types of all features.
- [] For numeric features: minimum, maximum, mean, median, standard deviation, quartiles, and range.
- [] For categorical or text features: frequency counts, unique values, and class distributions.
- [] Appropriate visualizations (e.g., histograms, boxplots, bar charts, density plots, or frequency plots).
- [] Present the results in tables and visualizations that clearly describe the structure and distribution of the dataset.
- [] (written analysis) **Normal vs. Non-normal distribution:** Using your own words, please describe which features have a normal vs. non-normal distribution and provide supporting evidence for this. The supporting evidence can be a data visualization, graph or table of the descriptive statistics.
- [] (written analysis) **Potential Impact of Normal vs. Non-normal distribution on AI Model Development:** Using your own words, please write how you think the distribution of the features in your dataset could affect an AI model trained on the data.

[] **Feature Engineering:**

[] **Proposed New Features:** Create at least two new features derived from existing dataset features. Describe:

- The source feature(s) used
- How the new feature was created
- The datatype of the new feature
- Why the new feature may be useful for the AI Model Development Scenario

[] **Evaluation of New Features:** Produce descriptive statistics for the newly created features and compare them to the original features. Discuss whether the engineered features provide additional information, reduce ambiguity, improve representation of important concepts, or better align with the project objective.

[] **Submission Memo Requirement:** Please include your datasheet for your LLM dataset and the information requested above in your submission memo, in a methodology/data section or the Appendix.

Part B. Tokenization and Embedding Experiments (7')

[] Tokenization & Embedding Experiments:

[] Choose a tokenizer or chunking strategy, and apply it to your dataset.

- You may use your own tokenization method or those included in the start PY files. Please explain your choice of tokenizer in your homework submission memo.

[] Choose an embedding model & create embeddings from your tokens.

- You may choose your own embedding model, or download Embedding Gemma, or use one of the toy embedding methods included in the start PY files. Please explain your choice of embedding model in your homework submission memo.

[] Measure the similarity of your embeddings.

- You may choose your own similarity/ distance metric, or those included in the start PY files. Please explain your choice of metric in your homework submission memo.

[] **Interpretation and AI Model Development Implications (written analysis):** Using your own words, explain how the results of your tokenization and embedding experiments could influence:

- Model performance
- Generalization
- Bias and representation of concepts
- Dataset quality
- Future model training and fine-tuning decisions for your project

Support your conclusions using evidence from your experiment results.

[] **Submission Memo Requirement:** Please include your experimental results in your submission memo, in an Appendix.

Part C. Research Plan & Analysis Memo (6')

For Part C, you will describe your research plan for the final project and write an analysis memo describing your results from Part A and Part B of this assignment (AS01).

In the memo, please describe one (minimum) to three (maximum) research questions, which relate to your organization scenario and can be answered through your data and LLM training and evaluation work for this class.

Requirements:

[1] Please write the memo in your own words and conduct the analysis manually, e.g. do not use AI assistance. The reason for this requirement is that it is important to practice the skills in this assignment, particularly the writing and analysis skills, without AI Assistance, because employers still expect employees to develop and grow these skills, in part so they have the discernment and capacity to detect when an AI Assistant's performance is low-quality or erroneous.

[2] Please meet and consult with your final project team to decide on the AI Model Development Scenario and research questions you will pursue for your final project. Your research questions can be similar or the same as those on your final project team, but **not** similar or the same as other final project teams.

[3] If you've used an external source of information to complete any part of this assignment, please cite the resource, using the appropriate citation format. These references can include academic, government and industry literature; software tools; AI models; AI tools; API documentation; other technical documentation; social media; news; blogs; github repositories.

Assignment Report Requirements:

Canvas submission: Upload a PDF or Google document version of your memo to Canvas.

Google Document Access: Please grant the teaching team access to the Google document before submitting the link to Canvas. This will enable us to provide feedback.

Grading

Part	Requirement:	Weight of Total Grade: 20%
		<i>Weight of Part:</i>
A	Organization Scenario & LLM Training Dataset Datasheet	7%
B	Token & Embedding Experiments	7%

C	Research Plan & Analysis Memo	6%
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Grade Rubrics	Share of Assignment
Accuracy: have you understood the question; e.g. have you followed the instructions correctly. Here, correctness refers to attention to detail in following the described steps, including the assignment policies and noted strictions on data and technology use, as well as coordinating with your final project team.	5%
Completeness: Have you provided all the requested information in your submission memo, including each subpart of the question or instruction?	5%
Creativity & Innovation: Does your organization scenario, research plan, and accompanying LLM training dataset and analysis reflect creativity and a proposal that addresses a novel or innovative research question or problem? How does your project differentiate itself?	35%
Analytics & Insights: Have you written a thorough, high-quality memo, using your own words, providing the requested information? In your analysis, do you arrive at meaningful and informative insights?	35%
Scientific Rigor: do you provide detailed documentation, allowing for a repeatable and explainable research plan and analysis? (e.g., Code quality, documentation, and reproducibility of the results)	20%

Due Date: Monday, September 14, 2026, 11:59 pm EST.

Assignment Policies:

Your work should reflect a deep engagement with the material and a nuanced understanding of the capabilities and limitations of AI language models.

All Course Policies specified in the syllabus apply to this homework. Make sure to thoroughly read and understand the course policies. Please email the teaching team any questions you have. We are happy to provide further clarification or discuss any concerns or questions.

AI Use/Assistance: Please disclose or cite all uses of LLM to assist with the preparation of the homework assignment, including using LLMs to generate software code, write or format or edit the homework submission document, conduct research or help answer questions. We ask this because it follows good professional and ethical practices.

References:

[1] <https://www.cisa.gov/news-events/news/traffic-light-protocol-tlp-definitions-and-usage>

[2] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. Datasheets for datasets. Commun. ACM 64, 12 (December 2021), 86–92. <https://doi.org/10.1145/3458723>